



A Residential Robot for Chinese Cupping Physiotherapy

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Abstract. Chinese cupping physiotherapy is integral to traditional Chinese medicine but is labor-intensive and requires expertise. Recent advancements in automating traditional Chinese therapies involve large mechanical systems with computer vision, which may not be practical for residential use. Our research focuses on developing residential robots for cupping therapy, specifically designed for Chinese elderly individuals. These robots utilize a specialized gripper integrated with a compact robotic arm featuring a 63.3-centimeter reach and a suitable base for home placement. Employing an STM32 controller with magnetic encoders, we establish closed-loop control for the stepper motor, ensuring precise grip control and smooth trajectory. Experimental comparisons between open-loop and closed-loop systems underscore the significance of the latter in preventing step losses in the 2nd-axis and 3rd-axis stepper motors. Additionally, we validate the acceleration of each axis during movement using exact DH parameters and MATLAB robot toolbox simulations, which could meet the necessary requirements for this innovative approach.

Keywords: Cupping therapy · STM32 · 3D Printing · SolidWorks · MATLAB

1 Introduction

The integration of industrial robots into manufacturing systems represents a noticeable upward trend within the field of engineering. With the objective of effecting a paradigm shift in production systems, there is a concerted effort in both academic research and practical applications to embrace digital technologies, especially within the emerging era of Industry 4.0. Central to this transformation is the adoption of advanced control systems and precision-driven components in robotic technology, enabling robots to execute a diverse range of commands originating from the user interface [1]. This facet displays significant potential, as robots endowed with these attributes offer substantial advantages. The heightened levels of productivity, performance capabilities, and potential for increased profitability make such robots suitable for extensive deployment in industrial, manufacturing, and even medical service sectors [2, 3].

In the current context, there is a growing recognition of the positive impacts of traditional Chinese medicine, traditional therapies, and herbal remedies on human health, leading to an increased public interest in traditional Chinese therapeutic practices [4,5]. However, the insufficient number of certified traditional Chinese medicine practitioners hinders the ability to meet the rising demand from the public. As a result, despite the increasing interest in adopting traditional Chinese therapeutic methods, only a minority have had the opportunity for hands-on experience with such therapies. Nevertheless, there are potential solutions to address this situation. The rapid iteration of advanced robotics and sophisticated control systems in contemporary settings has enabled the widespread integration of robots across various domains, contributing to enhanced operational efficiency. Simultaneously, based on the preliminary investigation conducted by the research team, survey participants display a discernible inclination toward the adoption of robotic applications within the medical sector. To a certain extent, this outcome highlights the feasibility of harnessing robots in the realm of traditional Chinese medicine therapeutic interventions.

Hence, our research team resolved to employ robotic systems for the restoration of cupping therapy, a prevalent and extensively utilized technique within traditional Chinese medicine physiotherapy. Our objective was to attain precise replication of the cupping procedure through a meticulously devised control system, while concurrently addressing and enhancing the safety aspects within home environments. This entailed addressing potential safety risks stemming from the conventional open-flame methods. Notably, the conceptualization and development of such a robotic prototype, hitherto unexplored in the domain of traditional Chinese medicine, possess considerable potential. Successful realization and subsequent application of this robotic prototype could substantially alleviate the labor-intensive requirements of cupping, a traditional Chinese medicine physical therapy, thereby fostering its wider acceptance among the general populace [6]. Consequently, our research team perceives this undertaking as being endowed with both forward-looking and experimental significance.

Traditional Chinese medicine therapy robots should have the advantages of safety, convenient and fast use, easy movement, and good results. Based on the aforementioned considerations, our research consortium has established the viability of substituting conventional manual techniques in traditional Chinese therapeutic interventions with high-precision robotic alternatives [7]. Once the need for cupping treatment is determined based on the needs, machines can be used for treatment [8]. To render these interventions suitable for domestic use, our design framework has undergone meticulous refinements. Our foremost priority has been the enhancement of safety. Fire-related deaths and injuries are an enduring global public issue [9]. Each year in the United States, 2900 people die in house fires and this issue can be prevented [10]. In order to avoid the occurrence of fires as much as possible. We have replaced traditional open-flame canisters, commonly employed in traditional Chinese fire cupping, with non-combustible gas canisters. This strategic transition precludes the potential hazards associated with improper handling [11]. Our research collective has substantiated the

feasibility of substituting traditional manual methods of traditional Chinese therapeutic interventions with high-precision robotic alternatives[12]. To render the intervention conducive for domestic use, our design framework has been subjected to nuanced refinements. Primarily, prioritizing safety, we have adopted non-combustible cups in lieu of conventional open-flame cans in traditional Chinese fire cupping, thus preempting the potential hazards associated with improper operation[13]. Subsequently, our devised robotic system autonomously undertakes the cupping therapeutic procedure, facilitating usability even by individuals without specialized training while maintaining commendable precision. Furthermore, our robotic system’s physical dimensions are calibrated to integrate within domestic spaces, aligning with spatial constraints seamlessly. This design optimization, characterized by a more compact and tightly interconnected structure compared to conventional robotic arm configurations, confers enhanced maneuverability within relatively confined spaces.

Designing a residential robot for cupping therapy has the potential to enhance the accessibility of traditional Chinese medicine practices, thereby facilitating the preservation and popularization of traditional therapeutic techniques among a wider population. The emergence of this robotic technology is in accordance with the prevailing trend of home healthcare and telemedicine. This innovative robot offers a viable solution for individuals seeking therapeutic interventions without having to physically attend a healthcare institution, thereby addressing the difficulties posed by circumstances such as a pandemic where face-to-face consultations are arduous. The collection of data during therapy sessions can serve as a valuable resource for conducting research on the efficacy of cupping therapy, its influence on diverse health conditions, and the establishment of evidence-based procedures. The progress in technology can be observed through the creation of a robot specifically designed for cupping therapy in residential settings. This innovative development not only contributes to the field of robotics but also promotes the advancement of human-robot interaction and assistive technology. The provision of personalized care by robots can lead to enhanced patient outcomes and improved quality of care, as they are capable of delivering consistent and tailored treatment.

2 The Implementation of Robot

The development of a robotic arm with an integrated gripper structure at its terminus entails a multifaceted process that requires a comprehensive understanding of various domains, including mechanical and electrical engineering. This section provides a detailed account of the sequential steps taken in creating a six-degree-of-freedom robotic arm equipped with a can gripper. The narrative begins by discussing the selection of appropriate materials and essential software to enable the robotic arm’s functionality. Subsequently, it delves into the conceptualization and assembly of the robotic arm. Furthermore, it explores the coding aspects associated with our robotic arm.

In an endeavor to streamline the intricacies of the cupping procedure and optimize spatial utilization, we have designed the robotic arm to exclusively

manage the complete cupping process. The pivotal mechanical element within this context is the principal arm. After a thorough assessment of prerequisites by our research collective, it was strategically determined that a six-degree-of-freedom robotic arm would best suit our objectives. This specific configuration provides enhanced motion capabilities with six degrees of freedom, allowing for a diverse range of trajectories. This flexibility facilitates precise positioning of the cupping apparatus, accommodating various anatomical orientations, and enhancing the system's adaptability to individual user needs. Additionally, innovative enhancements, particularly at the end effector of the robotic arm, are poised for integration to optimize the efficacy of the cupping procedure. A visual representation of our robotic model is depicted in Fig. 1.

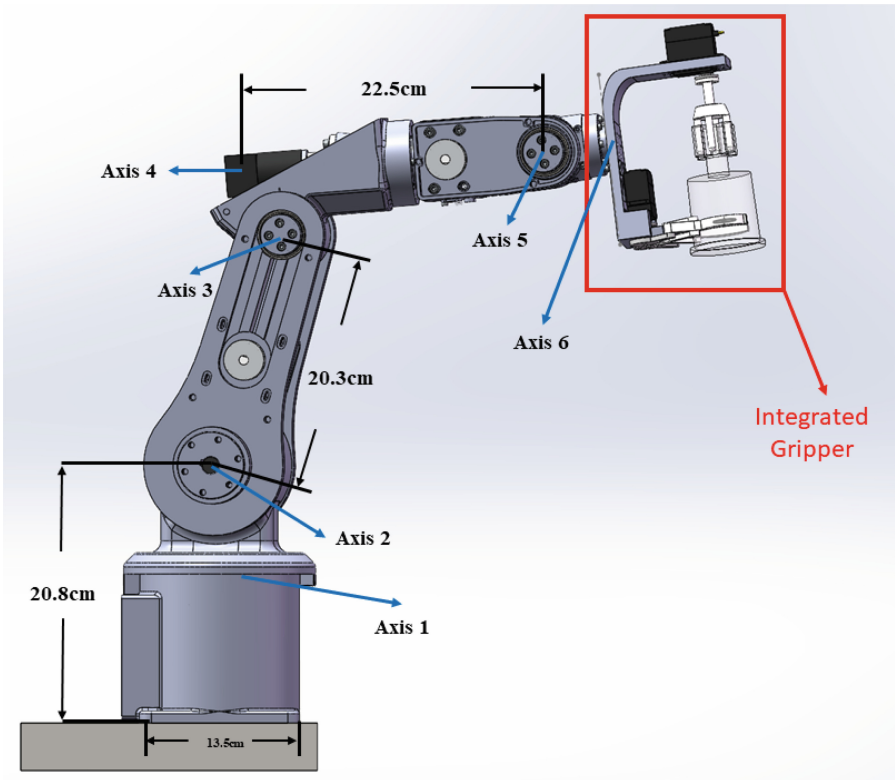


Fig. 1. The detail and corresponding data of cupping robot

2.1 Mechanical Design

In order to enhance the efficiency of gas canister extraction procedures and minimize the spatial requirements of the equipment, a singular robotic arm is employed for the complete process. The primary mechanical component in this

setup is represented by a solitary main mechanical arm. After conducting a preliminary assessment of the essential prerequisites, it was determined that a six-axis robotic arm should be conceptualized. This particular configuration bestows a significant degree of freedom in movement, thereby allowing for the execution of versatile trajectories. This design choice facilitates precise alignment of the gas canister to accommodate the diverse contours of users' bodies, thereby enhancing user adaptability. Furthermore, strategic improvements are incorporated at specific segments of the robotic arm, particularly at the end effector, with the aim of optimizing the efficiency of gas canister extraction.

The yaw axis of the robotic arm is controlled by a single motor. To prevent unwarranted arm movement that may result in wiring entanglement and safety hazards, mechanical constraints are introduced between the base and the first axis. This serves to curtail the range of motion, given that a full 360-degree rotation is not required during the canister extraction process.

At the juncture of the first axis, a force analysis reveals that the base primarily encounters pressures stemming from the upper arm and tangential friction induced by axis rotation. In such circumstances, a preferred approach involves the implementation of crossed roller bearings. These specialized bearings are engineered with rollers arranged in a 90-degree V-shaped groove, oriented perpendicular to each other. Nevertheless, due to the limitations inherent in standard models, a customized crossed roller bearing is devised. This distinctive design employs a sandwich structure, wherein rollers and lubrication are interposed between layers, culminating in a bespoke crossed roller bearing that aligns precisely with the project's requirements.

As we progress beyond the second axis, the extension of the arm leads to a repositioning of the arm's center. The second-axis motor plays a pivotal role in supporting the gravitational load exerted by the components situated above it. To mitigate the load-bearing constraints associated with conventional couplings, a keyed connection method is employed as an alternative to the traditional approach, thereby diminishing the risk of relative sliding.

For the third axis, motors are strategically positioned between the arms of the second axis, with synchronization gears facilitating the transmission of motion. This configuration not only alleviates the load on the second-axis motor but also ensures robust support for the third-axis motor, particularly during high-intensity operations.

The motors employed for the fourth, fifth, and sixth axes are compact and accompanied by optimized installation designs. The fourth and sixth axes are characterized by direct-drive configurations, while the fifth axis incorporates synchronous belts and gears. Crossed roller bearings, akin to the design implemented for the first axis, are seamlessly integrated into the fourth and sixth axes, contributing to enhanced overall performance.

The whole design of the gripper is based on a certain therapy cup shown in Fig. 2. The designed catching gripper has a thirty-degree range of motion, which enables it to accommodate the movement of cups through its capture region. Additionally, the rotation connection component features four columns

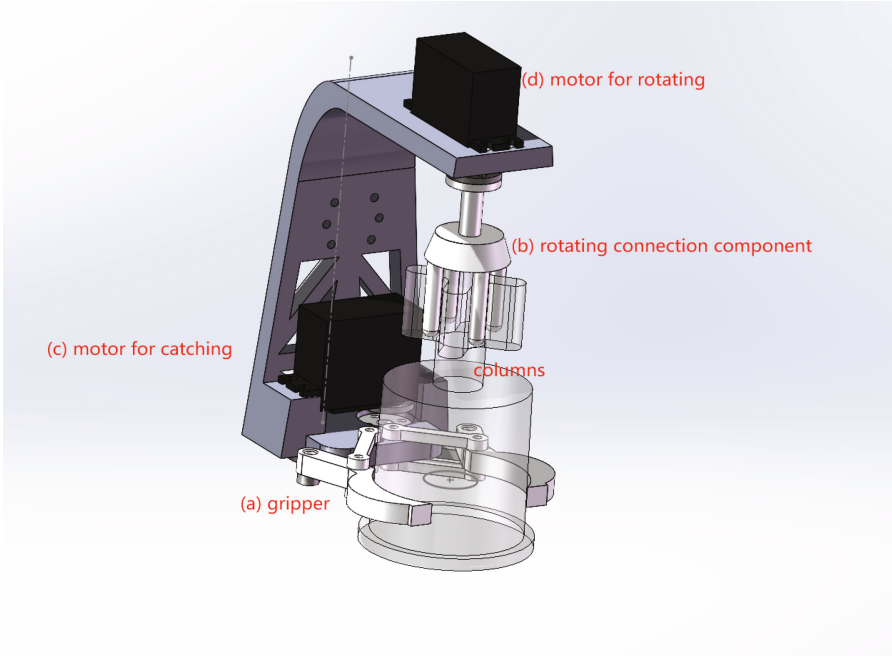


Fig. 2. The specially designed gripper for commonly used cup

at its base, which establish a secure connection with the upper part of the cup. These columns are intentionally inclined at an angle of approximately 5° towards the base to avert direct collapse during cup retrieval in practical implementation. Stepper motors, situated at positions (c) and (d), are employed for the specific tasks of cup grasping and subsequent rotation. Furthermore, we have implemented refinements aimed at reducing the weight at the rear section of the gripper. This design alteration incorporates void spaces without compromising the gripper's structural integrity.

2.2 Electrical Control System Design

In this project, the control of the whole system used an STM32f103C8T6 control board manufactured by JIENDI named “STM32F103CxPop”, with two sets of buttons manufactured by YwRobot named “four-button switch” that could interact with the user. It is noticeable that one button set consists of four buttons, while one set is for a user to control a robotic arm, the other is to control the gripper. Two MG996R servo motors are directly driven by the STM board, while 3 NEMA-15 stepper motors, 1 NEMA-23 stepper motor, and 2 NEMA-17 stepper motors are driven by driver boards integrated with the magnetic encoder. The details of stepper motors and magnetic encoder driver boards are shown in Table 1 and Table 2 respectively. Each button set consists of four buttons in the

color red, blue, yellow, and green, with the function of grabbing the cup, intermediate position, laying-position, and sitting-position cupping respectively for the first set. For the second set, the functions are to lock the cup, release the cup, rotate in a clockwise direction, and rotate in a counterclockwise direction.

Table 1. Details of stepping motors

Type	Static torque (N*cm)	Stepping angle (degrees)	Working current (A)	Reduction ratio	Highest rotational speed(RPM)	Weight (grams)
NEMA-17	28	1.8	1.5	50	800	410
NEMA-23	110	1.8	3	50	800	680
NEMA-15	5.1	1.8	0.6	1.2	800	200

Table 2. Details of driver board with magnetic encoder

Type	processor	Highest control speed (RPM)	Closed-loop control frequency (KHz)	Voltage required(V)	Current supply (mA)	Weight (grams)
MKS SERVO35C	32K ARM	2000	10	7-28	0-3000	410
ZDT x42	32K ARM	2200	20	7-28	0-3800	680
ZDT x57	32K ARM	4000	20	12-32	0-4000	200

In Fig. 3, a comprehensive circuit connection is depicted. Firstly, the power supply converting 220V into 24V connects to further DC-DC converter into 5V. The STM32 board with 5V voltage input further connects to Button sets, MG996R servo, and Magnetic encoder driver board. The connection of button sets and MG996R servo is through GPIO pins, while the power supply of these two components is 5V voltage connected to the previously mentioned DC-DC converter. It is noticeable that the connection between the magnetic encoder driver board and STM32 board is for communication only, with UART ports of STM32 through TTL to transmit information. Furthermore, all the serial communication of these driver boards is parallel connected between the TX and RX lines of UART ports of STM32. Secondly, the power supply of these driver boards is directly from the very first converter which is 24V, and the connection between the stepper motors and these boards is through HX2.54 for two windings power supply for motors. Lastly, the magnetic is equipped at the back of the motor axis to interact with module MT6816CT beneath the control board to encode magnetic information.

2.3 3D Printing

The assembly of the robotic arm is accomplished through the utilization of Ultimaker Cura, a 3D printing software employing the Fused Deposition Modeling (FDM) technique. The printer boasts dimensions of 330*240*300 mm, offering a substantial printing area for component production. Printing is executed at

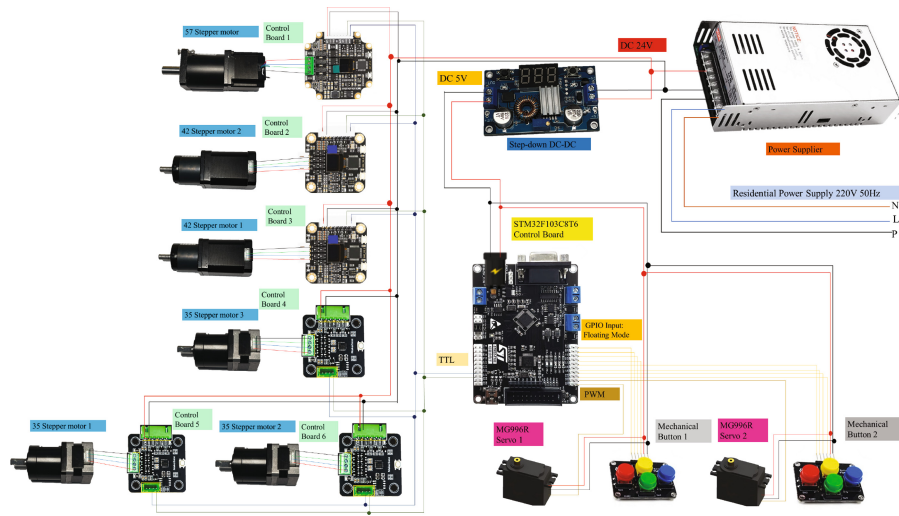


Fig. 3. The electrical control system design

a speed of 24 mm/s, ensuring high precision with an accuracy level of $6.9\text{ }\mu\text{m}$. This printer accommodates filaments of 1.75 mm diameter, including PLA, ABS, and TPU. For this particular print, we employed white polylactic acid (PLA filament). The printing process is completed in a single batch, which spans approximately 104 h. For visual reference, Fig. 4 (A) provides a depiction of the 3D-printed components comprising the 6-degree-of-freedom robotic arm, while Fig. 4(B) showcases the main body of our robot.

2.4 Close Loop System Design

The closed-loop control system for the stepper motor relies on the magnet positioned on the motor's rotor and a magnetic decoder located beneath the controlling driver board. Once the control command is transmitted from the STM32 to the Control driver board, the stepper motor is engaged by activating its two motor windings. Concurrently, the magnetic decoder reads the magnetic data and translates it into an accurate angle measurement. Moreover, the Control driver board compares the intended command destination with the current angle. In case the motor loses steps, the Control driver board compensates for the missed steps. Additionally, if the demanded torque significantly exceeds the stepper motor's capacity, meaning that the issued command is received but the decoder detects no movement in the motor's rotor, the Control driver board initiates a protective mode. This mode promptly halts rotation and places the stepper motor into an enabled state. In the initial iteration of our control design, an open-loop control system was employed, utilizing driver boards DRV8825 and TB660 to govern six stepper motors. Although experimental findings indicated that the stepper motor with a reduction ratio of 50:1 could accommodate the

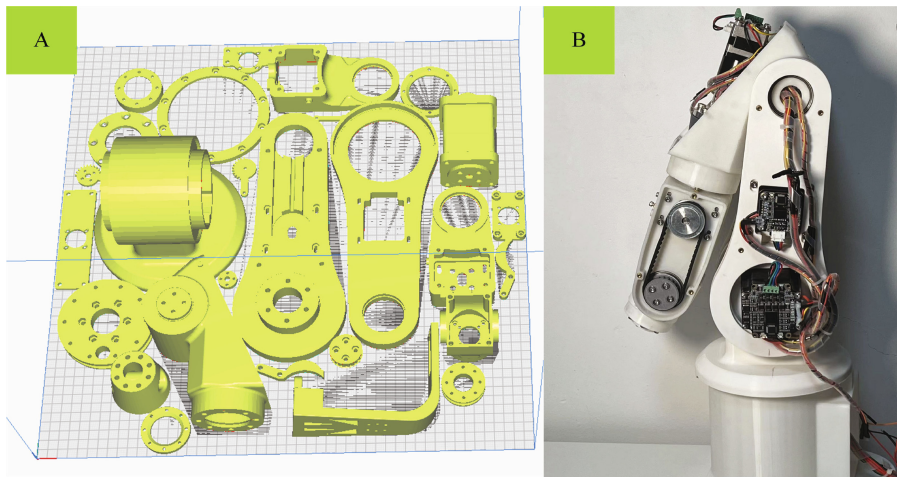


Fig. 4. The 3D Printing Components: (A) Slicing, (B) Assembled components

demands of our therapy load, it was observed that the precision of motor steps did not align with our commanded values. This discrepancy was found to be both common and unpredictable, rendering the system unsuitable for safe patient use. To substantiate these observations, we conducted data collection during testing, involving the meticulous measurement of angle variations and input commands from the controller. The further testing for closed-loop advantages is illustrated in Fig. 5. According to Fig. 5, the rotation state of the stepper motor using an open-loop drive without additional load is represented by the curve of test 1. Similarly, the rotation state of the stepper motor using a closed-loop drive without additional load is depicted by the curve of test 2. The curve of test 3 illustrates the rotation state of the stepper motor using an open-loop drive with an extra load. On the other hand, the curve of test 4 showcases the rotation state of the stepper motor using a closed-loop drive with an additional load. Moreover, the curve of test 5 demonstrates the rotation state of the stepper motor with a closed-loop drive when an artificial external force is applied. By comparing the data curves of test 1 and test 2, it is observed that in the scenario of an open-loop drive, the angle value experiences a sudden decrease at a certain moment followed by a normal increase. This phenomenon indicates that when a stepper motor is operated with an open-loop drive, there is a possibility of losing steps. However, when a closed-loop drive is employed, a magnetic encoder is installed, enabling the motor to constantly receive its current position and ensuring that no steps are lost. Consequently, closed-loop control enhances the stability of the robotic arm and enables it to achieve higher accuracy.

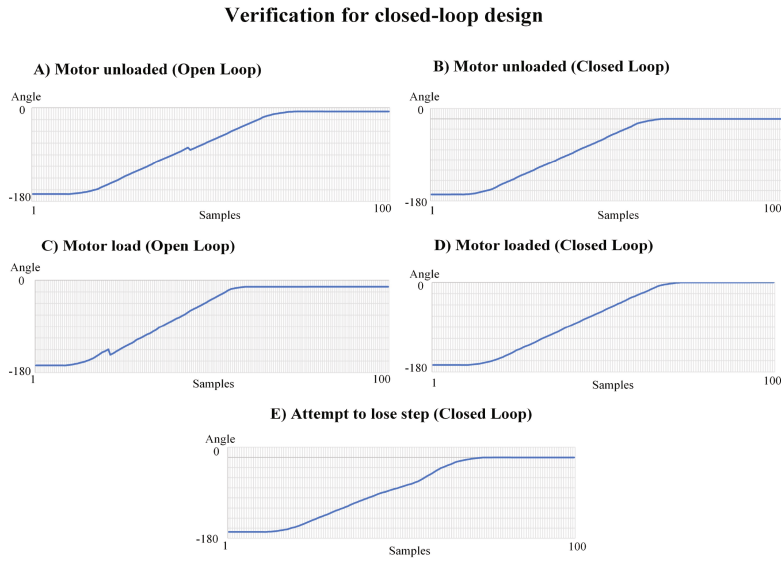


Fig. 5. Testing: (A) Test 1: Motor unloaded (Open Loop) (B) Test 2: Motor unloaded (Closed Loop) (C) Test 3: Motor load (Open Loop) (D) Test 4: Motor loaded (Closed Loop) (E) Test 5: Attempt to lose step (Closed Loop)

3 Experiment and Testing

3.1 Software Simulation

Table 3. THE BASIC DH PARAMETERS OF THE 6-DOF ROBOTIC ARM

Link	Theta	D	A	Alpha	Offset
Link 1	0	11.30	0	1.5710	0
Link 2	0	0	20.20	0	0
Link 3	0	0	6	1.5710	0
Link 4	0	16.40	0	1.5710	0
Link 5	0	0	0	1.5710	0
Link 6	0	2.6	0	0	0

Following the precise quantification of the actual dimensions and spatial intervals pertaining to each axis, the ensuing robotic parameters are presented as follows in Table 3. The simulation, implemented within the MATLAB Robotics Toolbox, utilizes the conventional Denavit-Hartenberg (DH) method to formulate six transformation matrices corresponding to the respective axes. These matrices are subsequently multiplied to derive the forward kinematic solution. The

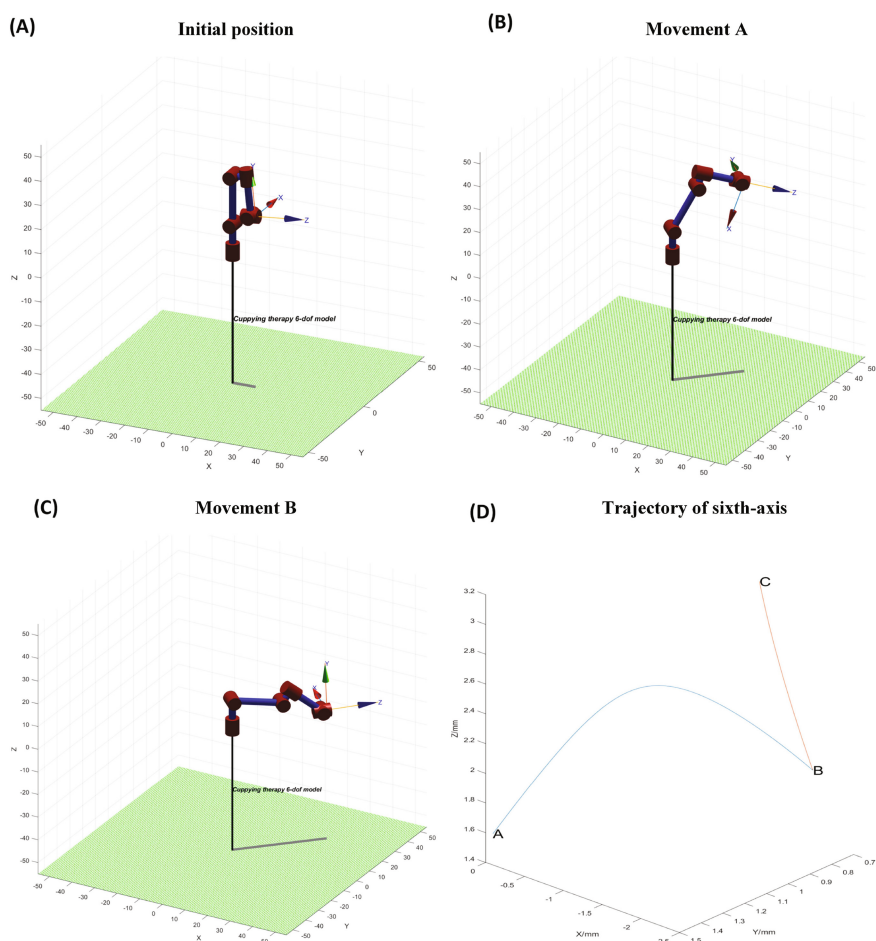


Fig. 6. The simulation of robot arm movement, (A) the initial position, (B) Movement A, (C) Movement B, (D) the trajectory of the sixth axis

inverse kinematic equations are obtained by further multiplication involving the inverse matrix operation while accounting for additional constraint conditions.

The cupping therapy movement includes two steps, firstly move from the initial position to the ready position, which could avoid collapse with the user's back, and second step is from the ready position to the final position. The inverse movement is the same with the changed sequence of the two steps above. The initial angle of the sixth axis must be ninety degrees, to save space for the gripper, When the machine goes to the ready place, the angle turns to zero to ensure the cup is perpendicular to the user's back, and the simulation and orbit (from A to B to C) is shown in Fig. 6. The further parameters of angles, velocities, accelerations in terms of time for initial to ready position then the

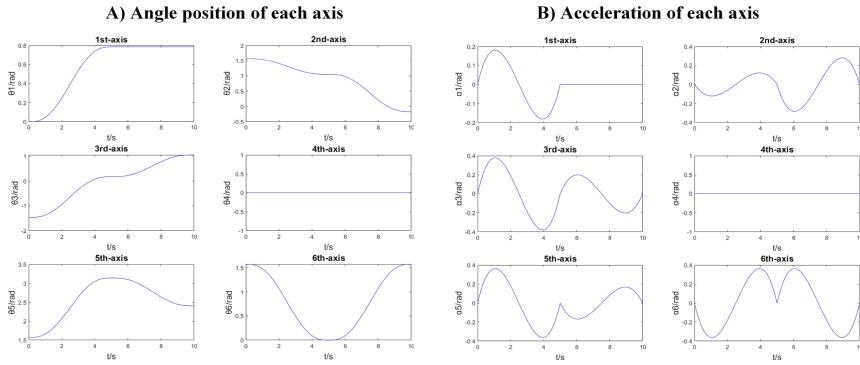


Fig. 7. The simulation result and comparison: (A) angle of each axis, (B) acceleration of each axis

therapy position movement is processed via MATLAB. The results are shown in Fig. 7 in terms of angle and acceleration respectively, and the experimental comparison result is in red and blue respectively shown in Fig. 8 for velocity.

It is evident that the simulation results are presented in Fig. 7(A), and Fig. 7(B), respectively. Through the specification of both the initial position and destination coordinates for our therapeutic procedure, the robotic arm exhibits autonomous acceleration along each axis. In contrast, owing to the command methodology employed by the driver board, the robotic arm under the control of our STM board experiences continuous acceleration, thereby yielding the disparate outcomes depicted in Fig. 8. It is worth noting that the velocity profile appears notably smoother in comparison to the actual experimental results. This discrepancy in trajectory, while stemming from identical starting and ending points, is substantiated by the shared integration, thereby affirming the subtle distinctions observed.

3.2 Testing

The experimental trials were conducted to validate the functionality of the constructed robotic arm. Its main tasks included successfully capturing a gas canister, transporting it to the designated location on the target's body, and utilizing a gripper to facilitate bidirectional rotation for both gas intake and release. This process aimed to regulate pressure. The upper segment of the robotic arm is extended to position the gripper accurately, with the arm's end being attached to the gripper itself. The lower gripper directly seized the gas canister and affixed it to the target's back. Subsequently, the canister underwent rotation to elevate the enclosed pad, creating an area of reduced pressure. Following a brief waiting period, the robotic arm reversed its rotation to augment the pressure within the canister. Once this step was completed, the robotic arm returned to its initial placement, bringing the canister back to its original position. Additionally, it's worth noting that the robotic arm successfully executed the task of working with

Comparison of velocity of each axis between simulation and experiment

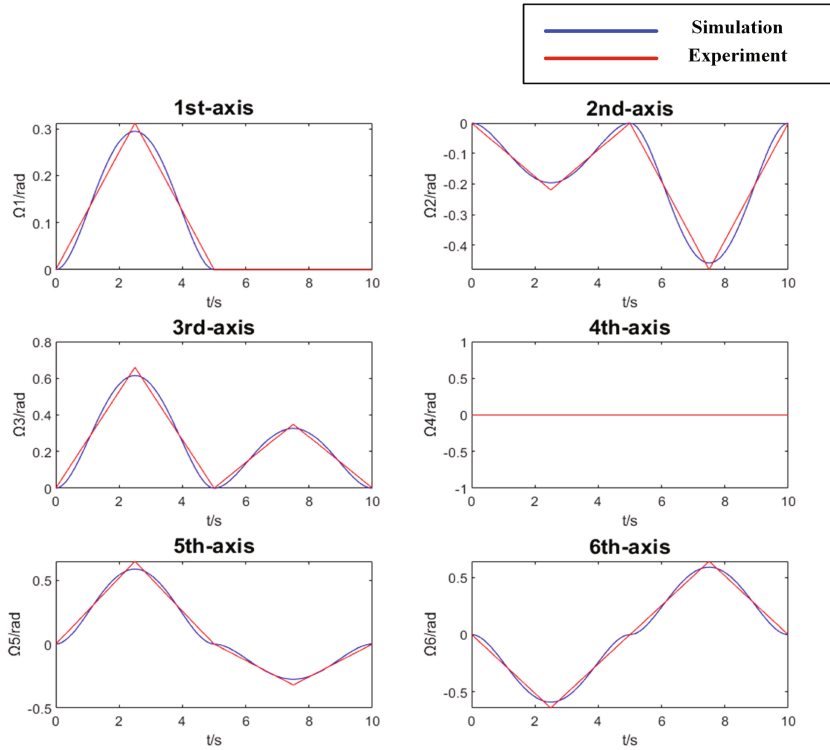


Fig. 8. The simulation result and comparison: Comparison of simulation and experiment in terms of velocity of each axis.

a canister in a sitting position. This scenario resembled the robotic arm situated on a table while the target subject sat on a chair. The robotic arm approached the initial location of the canister on the table. By rotating the fourth motor by 90° and the sixth motor by 60° , the working plane was adjusted parallel to the target's back. Consequently, the canister adhered to the target's back, enabling the rotation of the canister and the vertical movement of the pad, replicating the procedure used in the lying position. In summary, the experimental testing unequivocally demonstrated the efficiency of the robotic arm and its gripper in accomplishing the designated tasks.

4 Conclusion

In summary, this paper has undertaken the pioneering development and comprehensive analysis of a high-precision robotic system, with a primary emphasis on its application in gas can treatment interventions. Through the intricate integration of mechanical engineering, control systems, and advanced technological

components, this research has unveiled a transformative pathway within the field of therapeutic interventions.

The meticulously crafted six-axis robotic arm, blending principles of 3D printing and closed-loop control mechanisms, serves as a testament to the convergence of cutting-edge engineering. Its compact design, tailored to fit spatial constraints, and its precise maneuverability, represent a novel approach to enhancing therapeutic precision. The amalgamation of an STM32 main controller, microprocessors, and magnetic decoders facilitates seamless task execution, demonstrating the synergy of digital innovation and mechanical expertise.

Central to this study is the groundbreaking gripper, which epitomizes an unprecedented fusion of rotational capability and secure gas can capture. Empirical trials have underscored the system's accuracy and versatility, with a particular focus on the pivotal role of the gripper in capturing gas canisters while enabling bidirectional rotation. This gripping mechanism, in conjunction with the holistic robotic system, has the potential to significantly elevate the precision, efficacy, and accessibility of gas can treatment interventions, addressing a critical need in modern therapeutic practices.

The ramifications of this research extend far beyond the realms of robotics and engineering, permeating into the domain of healthcare delivery. The precision-driven approach of this system has the potential to redefine therapeutic interventions, enhancing the quality and effectiveness of treatments. As medical practices continue to evolve, the integration of robotics in healthcare holds promise for personalized, accurate, and efficient interventions tailored to meet individual patient needs.

In this context, this study constructs a compelling narrative at the intersection of advanced robotics and therapeutic applications. By showcasing the convergence of engineering ingenuity and healthcare demands, it underscores the transformative potential of robotic technologies in shaping the future of medical interventions. While this research has primarily focused on gas can treatment, its broader implications open avenues for further innovation and exploration, emphasizing the interdisciplinary nature of progress in both engineering and healthcare domains. As the field continues to evolve, this work stands as a foundational pillar, inspiring future researchers and practitioners to harness technology for redefining therapeutic precision and accessibility.

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